

Definition 3.2 (neighborhoods). As in the metric case, a *neighborhood* of a point x in a topological space is an open set containing it.

Definition 3.3 (closed sets). A set $F \subseteq X$ is *closed* if the complement $X \setminus F$ is open.

Corollary 3.2. *Finite unions of closed sets are closed, and arbitrary intersections of closed sets are closed.*

Proof. If $F_1, \dots, F_n$ are closed sets in X , then

$$X \setminus (F_1 \cup \dots \cup F_n) = (X \setminus F_1) \cap \dots \cap (X \setminus F_n)$$

is open, being the intersection of finitely many open sets (corollary 3.1), so $F_1 \cup \dots \cup F_n$ is closed.

If $\{F_\alpha\}_\alpha$ is any family of closed sets, then

$$X \setminus \bigcap_\alpha F_\alpha = \bigcup_\alpha (X \setminus F_\alpha)$$

is open, being the union of open sets, so $\bigcap_\alpha F_\alpha$ is closed. □